

A Method of Mental Care Using CNN, Transformer, TTS, Voice Conversion, and Whisper

서재범

Abstract—This essay discusses the potential of an AI-based mental care system. The proposed system is composed of emotion recognition using CNN, conversation generation with a Transformer-based LLM (GPT) and Google TTS, voice conversion using RVC V2, and input processing through STT (Whisper). The emotion recognition capability of CNN was tested using the FER-2013 dataset. As a result, the feasibility of system implementation was confirmed, but the experiments remained at a basic level. Tests on Transformer performance and voice conversion were also conducted on a limited scale.

I. INTRODUCTION

This essay explores the potential of a system designed to provide mental care to users using Convolutional Neural Networks (CNN), Transformer, Text-to-Speech (TTS), Voice Conversion, and Whisper. In summary, it investigates the feasibility of recognizing users' emotions by analyzing facial expressions with a shallow CNN using the FER-2013 dataset and suggests room for further development. It provides a technical basis for inducing appropriate responses through prompt engineering in a Transformer-based LLM. Additionally, it conducts a simple survey using audio generated from TTS and Voice Conversion models, and it presents the potential of low-latency speech recognition using the Whisper Speech-to-Text (STT) model. The limitations of this essay include the following: the CNN architecture was shallow, allowing only a basic exploration of feasibility; it does not include actual results of prompt engineering applied to the LLM; the survey comparing the naturalness of audio generated by TTS and Voice Conversion models was conducted on a small number of participants; and the STT tests with Whisper were performed using very short sentences, resulting in only basic experiments.

II. FACIAL RECOGNITION USING CNN

CNN is a relatively lighter model compared to the currently used Transformer and other models, making it more advantageous for low-latency image processing, where communication with the user is crucial. For basic experimentation, a simple model was designed, which will be referred to as SIMPLECNN in this essay. This architecture stacks 3 CNN blocks, extracts features from the image through a Fully Connected (FC) layer, and ultimately performs emotion classification.

The model structure is as follows:

CNNBlock: Each block consists of two Conv2D layers and a ReLU activation function, with padding applied to maintain the output size. SIMPLECNN: After stacking 3 CNN blocks (with 64, 128, and 256 channels, respectively), the output size

is fixed using AdaptiveAvgPool2d, followed by a Flatten layer to convert the output into a 1D vector. Then, 4 FC layers are sequentially connected, and finally, 7 classes are output. This model prevents overfitting through dropout, applying dynamic dropout to drop neurons with varying probabilities during training, improving the model's generalization performance. Therefore, the experiment was conducted using CNN-based facial recognition.

A. FER-2013 Dataset

The FER-2013 dataset includes seven emotion classes (Angry, Disgust, Fear, Happy, Sad, Surprise, Neutral) and is a standard dataset for facial expression recognition. The reason for selecting this dataset is that it contains a sufficiently trainable amount of data and consists of standardized images with a size of 48x48 pixels, enabling fast training.



Fig. 1.

B. Experimental Results

These are the results of the shallow CNN experiment using the FER-2013 dataset.

- Accuracy of Angry: 43.22%
- Accuracy of Disgust: 50.45%
- Accuracy of Fear: 43.46%
- Accuracy of Happy: 79.43%
- Accuracy of Sad: 48.99%

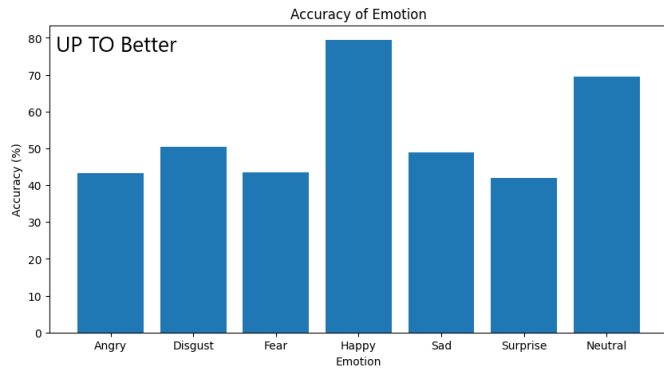


Fig. 2. CNN Experiment Results Using the FER-2013 Dataset

- Accuracy of Surprise: 41.86%
- Accuracy of Neutral: 69.43%

To verify technical feasibility, the FER-2013 (Facial Expression Recognition 2013) dataset was used. To apply it in a real environment, it would be necessary to augment the dataset or collect additional data.

The model used was a general CNN. The above graph shows the results. Due to time constraints, a shallow CNN model was created, but with more time, it is concluded that the recognition success rate could have been improved through the design of a more sophisticated CNN model.

The source code and dataset used to produce these results are attached in the references section with GitHub and Kaggle links.

It has been confirmed that recognizing users' emotions through facial recognition is technically feasible. Now, let's look into the Transformer-based LLM model for conversation generation.

III. INDUCING APPROPRIATE RESPONSES THROUGH TRANSFORMER-BASED LLM

The possibility of reading the user's facial expressions through facial recognition was confirmed above. Now, based on this, we aim to examine the technical foundation of GPT (Generative Pre-trained Transformer) and how it can be guided in a specific direction according to the situation.

A. Transformer?

The Transformer architecture was introduced in the 2017 paper "Attention Is All You Need" and is a structure that efficiently learns the relationships in sequential data and generates natural outputs through an understanding of time series. This paper demonstrated that by processing important information in parallel through the Self-Attention mechanism, it is possible to efficiently learn large-scale textual data and generate complex sentences.

B. GPT is a Transformer-based model.

By looking at the three papers, "Improving Language Understanding by Generative Pre-Training", "Language Models are Unsupervised Multitask Learners", and "Language Models are Few-Shot Learners", it can be said that GPT is a Transformer-based model. These papers directly mention "Attention Is All You Need", or include sentences stating that it was possible thanks to the Transformer architecture, or that the decoder of the Transformer architecture was used.

C. The Validity of Prompt Engineering

In the case of a happiness state, the rule "It seems like something happy has happened. Please celebrate together." could be applied. For example, if CNN predicts a state of sadness, the word "sadness" can be transformed into a sentence like "The user is feeling sad. Please consider this state and try to start a conversation." through appropriate mapping and delivered to the model. In the LLM model, it is possible to guide the conversation to start with content such as "You've been having a hard time lately, haven't you? Would you like to talk to me about it?" Another example would be if the user is in an "angry" state, "The user is currently angry. Listen to the reason for the anger and try to converse and empathize." could be sent according to the rules, and in the case of disgust, the rule could be "It seems something has happened. Ask about that issue and comfort them." In any case, if prompt engineering is possible, it is judged that it would also be able to provide positive responses. This is the judgment of this essay.

IV. TTS(TEXT TO SPEECH) VS. TTS AND VOICE CONVERSION

Given the importance of mental care for users, we determined that a more human-like, natural, and comforting voice is essential.

In this session, we use TTS and Voice Conversion models to transform voices and aim to draw a preliminary conclusion based on survey results regarding which voice feels closer to a human, is more suitable for providing comfort, and conveys emotions more effectively.

A. What is TTS?

TTS stands for Text To Speech, which is a speech synthesis model that converts text input into speech. For example, if you input "안녕하세요," it reads "안녕하세요" in an electronic voice. In this essay, we plan to proceed using B.wav (to be submitted along with the source files).

B. Voice Conversion?

A Voice Conversion model refers to a model that can convert the characteristics of a speaker's voice contained in audio into the voice of another person. For example, if the

audio output from TTS is input, it transforms it into a trained voice. Although not used in the survey, for an easy example of the conversion, the audio files EX01.wav and EX02.wav are attached to the current LaTeX source.

C. Using TTS Voice Conversion model.

The TTS used was the TTS available in Google Translate, and the Voice Conversion model utilized was based on RVC V2 with trained audio.

The reason for selecting the TTS used in Google Translate is that it is a familiar TTS that users have heard the most. RVC V2 was selected because it can learn a user's voice with approximately 10 minutes of audio and produce decent quality, making it suitable for basic research. The audio files EX01.wav and EX02.wav are attached to the LaTeX source.

D. TTS(Text To Speech) VS TTS -> Voice Conversion

Before Starting Initially, I wanted to include and discuss audio files in the essay. However, I determined that one of the original audio files used for training with the RVC V2 model is likely subject to copyright restrictions. Therefore, in this session, I will conclude by presenting the survey results without providing the audio files. Voice A is audio converted

TABLE I
TTS AND VOICE CONVERSION SURVEY RESULTS

질문	음성 A	음성 B
A 음성과 B 음성 중 어느 쪽이 더 사람에 더 가까운가?	10	6
A 음성과 B 음성 중 어느 쪽이 더 위로받기에 적합한가?	11	5
A 음성과 B 음성 중 어느 쪽이 감정을 더 잘 전달했나?	12	4

using the RVC model applied to TTS, while Voice B is audio directly generated from TTS without conversion. This survey was conducted with 16 participants using Naver Form. The survey evaluated which voice sounded closer to a human, which voice had better emotional expressiveness, and which voice was more suitable for providing comfort.

The participants were general users, representing a sample of typical user experiences. An additional limitation not mentioned in other sections is that the survey involved only a small number of participants and speakers of a specific language, raising questions about its true representativeness. If given the opportunity to write this essay again, I would aim to conduct research involving a larger and more diverse group of users.

The evaluation method involved choosing between Voice A and Voice B. The survey results are recorded in Naver Form, and the detailed form content is attached in the references section with a link to Naver Form.

The results showed that Voice A was superior in all survey categories. Based on this, we reached a preliminary conclusion that training the model with a moderately ideal

voice and converting the audio generated by TTS can have a more positive impact on users' psychology.

V. IMPLEMENTING LOW-LATENCY STT USING WHISPER

Not only speaking to the user but also listening to the user's voice and responding is considered important as a form of mental care. Therefore, I aim to explore the possibility of using Whisper, the most well-known STT model, to convert speech to text with low latency.

A. 1.The Technical Foundation of Whisper.

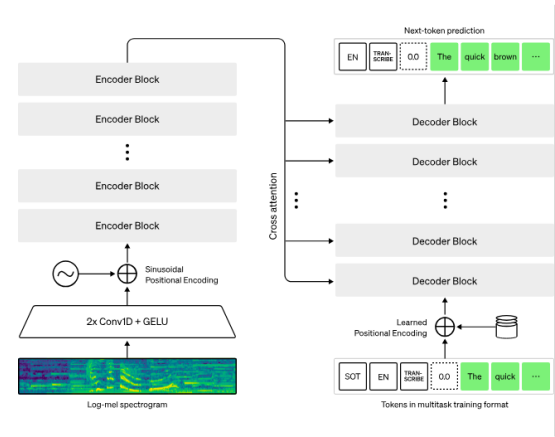


Fig. 3. Whisper model

It is a speech recognition model developed by OpenAI, operating based on Transformers. It has been trained on a vast amount of speech data and can be used for tasks such as recognizing speech in multiple languages, translating, and converting text. The model can take audio input with a 16kHz sampling rate and convert it into text.

The image above is provided to demonstrate that the model is built with Transformers. The use of encoder, decoder, and cross-attention structures serves as evidence that it is based on the Transformer architecture.

B. Possibility of implementing low-latency voice processing.

Low-Latency voice processing. While I could not upload Voice A used above, Voice B was generated using the TTS functionality of Google Translate.

I have uploaded Voice B along with an example of generating text using code. This can be checked on the GitHub link below. When converting Voice B through the Whisper model

```

import whisper
# 음성 시간: 0.3966745948791504 seconds

import time
start = time.time() # 시작 시간 저장
# Whisper 모델 로드
model = whisper.load_model("tiny") # "tiny", "base", "small", "medium", "large" 중 선택 가능

# 음성 파일 변환
result = model.transcribe("0.wav") # "0.wav"는 변환할 음성 이름
finish = time.time() - start # finish: Unknown word.
# 출력 결과
print("Text:", result["Text"])
print("Time:", finish, "seconds") # finish: Unknown word.

```

Fig. 4. CODE

```

Text: I'm here for you. Do you want to talk about it?
Time: 0.39366745948791504 seconds

```

Fig. 5. Conversion result and time

C. Conversion result

The above results were obtained. It took less than 1 second to convert the given sentence into text. Therefore, it is judged that low-latency STT can be implemented with appropriate logic and input devices (such as microphones).

VI. OVERVIEW OF THE INTEGRATED MODEL

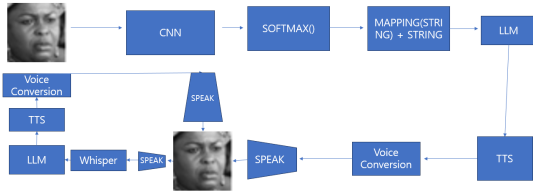


Fig. 6. Diagram of Final Integrated Model

CNN observes the user's facial state to extract features and analyzes facial expressions through SoftMax. It calculates the probability of the most likely state using SoftMax and proceeds with post-processing.

The derived probability values are combined with predefined mapping rules and strings to be converted into appropriate input and then delivered to the LLM (GPT).

The LLM (GPT) generates an appropriate sentence based on predefined prompt engineering. This generated sentence is then passed to the TTS (Text-to-Speech).

The TTS converts the received sentence into speech and delivers it to RVC V2 (Voice Conversion model). RVC V2 converts the speech into the user's voice and plays it back to the user (Speak).

The user listens to the converted speech and responds with their voice. The system keeps the microphone open for 10 seconds, waits for the voice input, and converts it into text using Whisper, which is then sent to the LLM (GPT).

The LLM (GPT) considers the previous conversation context (including prompt engineering) to generate a new sentence, which is passed to the TTS.

The TTS converts the generated sentence into speech and delivers it to RVC V2 (Voice Conversion).

RVC V2 converts the speech and plays it back to the user.

This process is repeated, allowing the system and the user to continue their conversation. It is expected that the user will experience mental care through this conversational process.

VII. CONCLUSION

This essay explored the potential of a mental care system using CNN, Transformer-based large language models (LLM), Text-to-Speech (TTS), Voice Conversion (RVC V2), and Whisper.

CNN demonstrated the possibility of real-time processing by capturing facial expressions based on the FER-2013 dataset. The Transformer-based LLM confirmed its technical feasibility for facilitating natural conversations with users through prompt engineering, based on emotion analysis results. Through a voice comparison experiment using Google TTS and RVC V2-based Voice Conversion, it was roughly proven via surveys that converted voices have better emotional delivery capabilities to users. Whisper confirmed its technical feasibility for real-time speech-to-text conversion using a 16kHz sampling rate and Transformer architecture.

This system could contribute to psychological stability through more natural interactions with users. However, this essay conducted

experiments using a limited dataset and a simple CNN model, and the small number of survey participants made it difficult to provide a complete evaluation of TTS, Voice Conversion, and STT

If, in the future, an environment is provided where this topic can be the sole focus without other commitments, more well-designed models, larger datasets, and feedback from more test users could be used to apply the system in real-world scenarios, leading to more detailed evaluations and conclusions.

VIII. REFERENCES

- **Prompt Engineering (LLM Response Guidance):** IBM Prompt Engineering
- **GPT Prompt Engineering Guide:** Prompting Guide
- **Evidence that GPT is Built on Transformer Models:** Attention Is All You Need (arXiv)
- **Voice Differences Using TTS and RVC Models:**
 - **Google TTS:** Google Translate TTS
 - **RVC GitHub Link:** RVC Project on GitHub
 - **Zero-Shot Voice Conversion Paper:** Zero-shot Voice Conversion (arXiv)
- **Evidence Supporting Whisper's STT Capability:**
 - **Whisper GitHub:** OpenAI Whisper GitHub
 - **Whisper Official Site:** OpenAI Whisper Official Site
 - **Evidence that Whisper Resamples to 16kHz:** Whisper Discussion on GitHub
- **Resources on CNN and FER-2013:**

- **CNN Paper:** CNN (arXiv)
 - **FER-2013 Dataset:** FER-2013 Dataset on Kaggle
- **GitHub with CNN and Whisper Code:** SION001122
GitHub Repository
- **GPT-1:** Improving Language Understanding by Generative Pre-Training
- **GPT-2:** Language Models are Unsupervised Multitask Learners
- **GPT-3:** Language Models are Few-Shot Learners
- **Hopes for Mental Care:** Positive Thinking